Course Name: Computer Networks and Internet Protocol

Assignment 4 - Week 4 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1

Let the maximum TCP payload be given 1450 Bytes; then what is the maximum network layer payload in Bytes?

a) 1450 Bytes
b) 1470 Bytes
c) 1430 Bytes
d) 1480 Bytes

Correct Answer: (b)
Detailed Solution: The TCP header size is 20 Bytes. So, the maximum network layer payload is (Max TCP payload + TCP header size)=1450+20=1470 Bytes.

QUESTION 2

In a distributed system, a server advertises a receiver window of 0 due to lack of buffer space. However, the application frees up space, but the updated acknowledgment is lost. What should the client do to prevent deadlock?

a) Wait indefinitely until the server sends a new acknowledgment.
b) Resend data packets at periodic intervals to elicit a response.
c) Use a keep-alive mechanism to periodically check the buffer status.
d) Close the connection and restart communication.

Correct Answer: (c)
Detailed Solution: A keep-alive mechanism allows the client to periodically probe the server for updated buffer availability without retransmitting data, preventing deadlock.

QUESTION 3

Which of the following statement(s) is/are true for the Transport Layer?

(i) The transport layer protocol should be Stateful.
(ii) Uses FTP protocols for ensuring flow control.

a) Only (i)
b) Only (ii)
c) Both (i) and (ii)
d) None of the above

Correct Answer: (a)
Detailed Solution: The transport layer needs to remember the state of the pipe so that appropriate actions can be taken. We need a stateful protocol for the transport layer. Moreover, ARQ protocols are used to ensure flow control.

QUESTION 4

A TCP connection is experiencing high delays and packet reordering. Which features of TCP help maintain reliable and ordered communication in this scenario?

(i) Sequence numbers in the TCP header.
(ii) Acknowledgment numbers indicating the next expected byte.
(iii) The urgent pointer to prioritize packets.
(iv) Sliding window flow control.

a) (i), (iii), (iv)
b) (i), (ii), (iii)
c) (ii), (iii), (iv)
d) (i), (ii), (iv)

Correct Answer: (d)
Detailed Solution :
A and B: Sequence and acknowledgment numbers allow TCP to reorder out-of-order packets and ensure reliable delivery.

D: Sliding window flow control ensures efficient data transfer despite delays.

C: The urgent pointer is unrelated to packet reordering or delays

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QUESTION 5

In the TCP connection release process, why does the active closer enter the TIME-WAIT state after receiving the final acknowledgment?

a) To confirm that the other host has received the FIN message.
b) To allow the passive closer to resend data if needed.
c) To ensure that any delayed packets in the network are discarded.
d) To avoid reusing the same port for a new connection.

Correct Answer: (c)

Detailed Solution: The TIME-WAIT state ensures that all delayed packets in the network from the just-closed connection are discarded. This prevents any potential interference with subsequent connections that might reuse the same port and sequence numbers.

QUESTION 6

Consider a network with a link bandwidth of 1 Mbps, a delay of 1ms, and a segment size of 1 KB (1024 bytes). Which of the following relationships is between BDP and segment size?

a) BDP = 8 x (segment size)
b) BDP/4 = segment size
c) BDP/8 = segment size
d) BDP x 8 = segment size

Correct Answer: (d)
Detailed Solution:
BDP = 1 Mbps x 1 ms = 1 Kb;
segment size = 1 KB = 8 x 1 Kb = 8 x BDP.

QUESTION 7

The use of a cryptographic function to generate sequence numbers for TCP connection is helpful to avoid

a) TCP Congestion Overflow
b) DoS attack
c) TCP SYN flood attack
d) None of these

Correct Answer: (c)
Detailed Solution: TCP SYN flood attacks can be prevented by generating sequence numbers using cryptographic functions.

QUESTION 8

A transport-layer protocol using AIMD starts with an initial sending rate of 1 Mbps. After four successful additive increases (each adding 0.5 Mbps), it detects packet loss. If the multiplicative decrease factor is 0.5, what will be the new sending rate?

a) 2.5 Mbps
b) 1 Mbps
c) 3 Mbps

d) 1.5 Mbps

Correct Answer: (d)

Detailed Solution :
Initial rate: 1 Mbps.
Four additive increases: $1+4\times0.5=3$ Mbps
Multiplicative decrease: $3\times0.5=1.5$ Mbps

QUESTION 9

Which of the following conditions does not influence the segment size in TCP?

a) Maximum Transmission Unit (MTU) of the underlying link.
b) Receiver's advertised window size.
c) Network congestion window size.
d) Header length of the IP protocol.

Correct Answer: (d)

Detailed solution: The Header Length of the IP Protocol does not directly affect the TCP segment size. Instead, it impacts the total size of the network packet but is unrelated to the TCP segmentation process.

QUESTION 10

In the case of a simultaneous TCP open (both hosts initiate a connection at the same time), which of the following are true?
(i) Both hosts will transition to the SYN-RECEIVED state after exchanging SYN messages.
(ii) The three-way handshake completes with the exchange of SYN+ACK messages.
(iii) Both hosts will immediately move to the ESTABLISHED state after receiving SYN messages.
(iv) Simultaneous open is less secure due to lack of proper sequence number synchronization.

a) (i) and (ii)
b) (iii) and (iv)
c) All are True
d) All are False

Correct Answer: (a)

Detailed Solution: In a simultaneous open, both hosts send SYN packets and transition to the SYN-RECEIVED state. The handshake proceeds as normal, ensuring proper synchronization of sequence numbers. Security is not compromised (D) because sequence numbers are negotiated during the handshake. So only (i) and (ii) are true.